Defining places for interchange: new tactial urbanist mobilityHubTools

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Abstract

1. Introduction

'Mobility hub' is a relatively new term for locations where a range of transport modes, especially transit and shared micro-mobility modes, and sometimes other services and facilities are offered. These are often developed to provide a range of non-private-and-petrol-driven-car alternatives to the surrounding communities and to interchanging travellers and as part of efforts to shift towards more sustainable travel behaviours and transport system outcomes. For many househouses having nearby access to shared electric cars, bikes and cargo-bikes, e-bikes and e-cargo bikes, or other vehicles might help in shifting towards more walking, cycling and transit-based travel through providing alternatives for those occassional trips that are otherwise difficult, such as if travelling with luggage, longer distances or at a particular time of day. Such households might be able to forgo purchase of a car or an additional car, or hasten their transition to use of electrically-powered modes that might have lower environmental impacts without the need to purchase an e-car themselves.

However, what makes a location a 'mobility hub', rather than just some other place where there are a range of travel options in close proximity, is unclear. Some mobility hubs, such as those in the Greater Toronto and Hamilton Area (GTHA) in Canada, are defined as such by a regional transport authority, with such status having implications for landuse planning and other institutional decision-making Salsberg et al. (2010); Engel-Yan and Leonard (2012). Others are created afresh, through some agency implementing facilities for a range of modes and/or other offerings in close proximity to each other and branding as a 'mobility hub' or some other similar term. In some cases, it may be that a range of offerings are already in place, especially where the private sector has already implemented car-, scooter-, bike- or other -share services, and converting somewhere into a 'mobility hub' is within reach by just giving the location a mobility-hub-related name in a press-release or on a website, and perhaps some minimal physical interventions such as signage and advertising.

Two problems, therefore, appear relevant: (1) that it is unclear what minimum characteristics a location might need to become a 'mobility hub'; and (2) that there may not be sufficient tools available to facilitate the transformation of such locations into 'mobility hubs' by either practitioners or, potentially through tactical urbanism and/or grassroots activism, the general public. This paper reports case and design-based research seeking to address these interlinked problems. The research adopts the Double Diamond Design process by iterating through alternative phases of diverging and coverging thinking related to discovery, definition, development and delivery UK Design Council (undated), with the discovery phase guided by

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exploratory field visits to a range of case study mobility hubs. A new package of software tools comprises the key research outcomes of the delivery phase, with these providing a range of adaptable and standardise mapping, branding and other outputs that might be used to temporarily implement a 'mobility hub', or as the basis for more permanent signage, branding and other physical installations. The rest of this paper is structured as follows: the next section outlines the context for theis research, including existing knowledge related to mobility hubs and tactical urbanism. The research methodology is then described, followed by presentation of the results. This are then discussed prior to a brief conclusion that includes the identification of directions for future research.

2. Research context

2.1. Mobility hubs

Mobility hubs are sometimes described as part of an emerging approach where shared and micro-mobility offerings might be linked, often with transit, to provide a range of options that support more sustainable travel. Early use of the term "mobility hub" occured in Toronto, Canada, as part of landuse and transport planning undertaken by Metrolinx and other governmental institutions across the GTHA and surrounds Salsberg et al. (2010); Engel-Yan and Leonard (2012). In "The Big Move" Metrolinx (2008) various transit stations and major interchanges were defined as "mobility hubs" for the purposes of supporting regional travel, modal interchange, and local connections including through transitions to more transit supportive landuse patterns in nearby areas.

The concept of having a range of transport modes, especially those that are shared or that involve micro-mobility (e.g. e-scooters, share bikes), geographically collected into a 'hub' may also have come about from the plethora of independent private-sector transport offerings that have recently emerged, including car-share and dockless bike/scooter systems. Although driven largely by developments in the start-up, technology and communications sectors, various governmental agencies have sort to both regulate such offerings, but also make them act as more than the sum of their parts, through defining dockless parking locations, allocating dedicated car parking spaces and other interventions, including through branding and informational signage that provide geo-spatial context and collectivisation. As well, there have been some emergingingproblems with the various generations of dockless bikes, e-scooters and e-bikes, including with respect to inappropriate parking of vehicles (i.e. blocking footpaths) and the fire safety and environment impacts of recaharging models that rely on batteries being collected in vans or small trucks and then charged, often in residential or other properties without the appropriate facilities to deal with electrical loads or battery fires Meigs (2023); Zhang et al. (2022). Shared electric mobility hubs, such as those currently being trialled in Ireland as part of the ROBUST project TRIP - Centre for Transport Research and Innovation for People (2025), may provide solutions to such problems by shifting the recharging of vehicles to be at a hub itself, which includes the appropriate electrical connections and equipment that, importantly for fires involving lithium batteries, are outside and typically unenclosed.

Some challenges, however, for implementers and communities may relate to the difficulty of developing mobility hubs, including the planning processes and funding acquistion necessary to build new facilities. Of course, there may be nothing especially new, other than the name, about mobility hubs given that railway stations have existing for over a century. Often containing platforms for boarding and alighting, waiting areas and information, cafes, parking lots, interchange with other modes such as buses, and many other facilities the 19th, 20th and current iterations of railway stations would appear to have largely already 'solved' for the problem of providing support for mobility at a network hub. However, railway stations tend to be located on land owned or otherwise controlled by railway companies or agencies. As such, facilities, signage, branding, information displays and most other aspects of the station environment might be largely under the control of a single institution, or at least a small number of closely-related agencies. Mobility hubs, in contrast, might tend to be located on public road reserves, within parking facilities or otherwise where there may be less clear authority and power to make change without widespread approvals or buy-in. These may also often require outside expertise and/or involve top-down policy-making in which those making decisions might be somewhat removed from the real mobility needs of individual communities or may adopt 'one size fits all' approaches that might lack responsiveness to local context. Public policy analysis has long since focused on such non-rational factors and how they might influence decision-making, but these tend to be overlooked in transport policy research Marsden and Reardon (2017), despite the applicability of instititionalism, political approaches, incrementalism and other models to transport Reynolds et al. (2017). Important within these are the top-down, bottom-up and hybrid theories of implementation, which emphasise the influence that street-level bureaucrats can wield to obtain outcomes that respond to their local needs and context, including sometimes outcomes that might be in opposition to central policy directions. Further exploration of this topic may be beyond the scope of this paper, but the extensive body of research related to the Advocacy Coalition Framework (including Sabatier (1986, 1987, 1988); Sabatier and Jenkins-Smith (1993); Weible et al. (2009); Weible and Sabatier (2007)) provides insights and theory. Perhaps of more relevance, however, is the field of tactical urbanism, which adopts many of the bottom-up and grassroots approaches to defining and re-imaging public spaces, especially those within road reserves and related to transport systems.

2.2. Tactical urbanism

Tactical urbanism refers to a range of approaches, often unsactioned, for making changes to public environments. These often seek to question what urban space or is for or the legitimacy of how it is allocated, such as in the case of "PARKing Day". This started as lunchtime take over of an on-street car parking space by an architecture firm, who rolled out some green grass and a park bench, paid the parking meter, and in doing so questioned the validity of using such spaces for storing vehicles instead of as a paklet for people. "PARKing Day" is now an annual event seeking, like the World Naked Bike Ride and Critical Mass, to demonstrate the possibility of change and question the legitimacy of the (car-dominated) status quo Lydon et al. (2012); Davidson (2013); Lydon and Garcia (2015); Blumgart (2016); Ride (2009); Morhayim (2012); Longhurst (2015). Similar 'pop-up' or protest-based approaches have helped legitimise bicycle lanes, delegitimise car-use of bus lanes, and are starting to becomem more mainstream (amongst engineers and planners) ways to help build public and institutional support for new transit lanes Fucoloro (2013); News (2019); TransitCenter (2018); of Transportation Studies (2019). These build on or have similarities to various earlier approaches including street-reclaiming Dobson (undated); Engwicht (1999); Fotel (2009); Institute (2015), the invention of the first 'woornef' Guttenberg (1982) and Local Area Traffic Management Damen and Millican (2017), as well as being evident in the make up of current directions towards 'Movement and Place', 'Complete Streets' and Network Operation Planning approaches Delbosc et al. (2018).

Despite the many names, variations and focii of these and other invterventional and decision-making approaches, the call from Marsden and Reardon (2017) to "pay greater attention to power. . . " retains relevance. Various authors have highlighted the challenges and poptential hazards of participating in unsanctioned activities, especially for minorities or others who might attract higher levels of public surveillance or opposition, or attention from authorities, for engagement in tactical urbanism or similar activities (INSERT CITATION HERE ABOUT PARK BENCH). Intersections between illegality, illegitimacy and the ill-enforced similarly exist in the context of street art and its place on a contiunum that includes commissioned murals, counter-culturism and the avent guard, graffiti, and all the way down to plain old boring vandalism. Kil (2017); Killen (2021) explore these topics in the context of limited operational and maintence impacts to railways, specifically trains. This, and larger questions related to 'what is art' are again beyond the scope of this paper, but not those that relate to who is allowed to post a public notice, define a 'mobility hub' or share knowledge about transport options, local facilities and services, and connecting modes.

Engineers, planners, operational managers, communications teams and others within transport agencies and governmental instutitions typically have less need to engage in tactical-urbanist-type implementation, typically because they have legitimate authority to make changes to the built environment, at least within the scope of their remit. However, pop-ups, trials, bottom-up and incremental approaches to implementation may still be of use, or even necessary, for such actors seeking to legitimise new, unusual or just controversial interventions, or if power is vested in political representatives or otherwise. Reynolds (2020); Reynolds and Currie (2021) defines these as 'pragmatic strategies for legitimation *through* implementation' based on case research of such approaches helping in the implementation of Bus Rapid Transit in Curitiba and other forms of transit priority in Zurich, Melbourne and Toronto. Other strategies include building legitimacy before implementation using technical evaluation, vision-based transport planning and/or public decision-making processes; or by avoiding impacts on those who might seek to delegitimise change.

Such public processes and technical activities are readily apparent in mobility hub implementation, as will be discussed in relation to some of the individual case examples that are discussed later in this paper. However, basic signage, linemarking or the public display of other information would appear likely to be relatively easy for many practitioners to implement within their own institutional setting. What may be lacking, however, are tools and standarised approaches to mobility hub definition and informational materials suitable for quick and relatively easy adaptation to the local contex of an existing mobility hub, or to 'create' a mobility hub out of various closely located modes, facilities and services.

Likewise, for activists, practitioners wishing to affect change outside the remit of just the confines of their specific employment suituation, or the general public there do not appear to be tools that might support tactical urbanism efforts related to mobility hubs. This might be particularly relevant for the 'creation' of a mobility hub out of already existing and geographically-proximate facilties, which might not be immediately obvious to travellers but something well-known and readily apparent for those with local knowledge of the geography and community. Leaflets, posters for shop windows and community noticeboards, or even the resources for a tactical urbanist to launch a 'guerilla marketing' campaign of posters, DIY signage and websites, and likewise might be all that is needed to facilitiate bottom-up action and empower communities to define their own hubs and advertise their pre-existing mobility services and related facilities. What appears to be lacking in the research literature, or in practice, are the tools to produce the sorts of maps, analysis and interpretations that might support the definition of a place as a hub and the communciation of its existing and potential mobility and other offerings.

As discussed later in this paper, existing mapping, signage and similar interventions in both the physical and virtual might tend to be produced on a bespoke (ie. location by location) basis by specialist geographers and graphic designers, and involve commercially-controlled datasets or platforms. With the advent of crowd-sourcing, open data and the widespread availability of open source software, however, there is the possibilty of developing location-bespoke, but generalised, mapping and similar tools and that might be useful for practitioners or others wishing to 'mobility hub-erise' some part of the urban environment. The methods used in this research to firstly define what such tools might do, and then to deliver a software package that enables others to more easily develop their own mobility hub mapping and informational materials, which might provide an initial version for future refinement, are therefore descrbed in the following.

3. Methodology

Coxon et al. (2018, 2021) outline various industrial design approaches and tools, as well as discussing their relevance and potential applicability to transport systems. Notable amongst these is the Double Diamond, which involves iteration across four phases, alternating between diverging discovery and then converging definition phases follwed by diverging development and convergance to a delivered design output UK Design Council (undated). This study adopts this approach as a framework to guide the research activity, which makes use of case studies of existing mobility hubs for the initial discovery phase.

Case studies can be especially effective for research into areas when theory development is at an early stage and when there is a need to explore phenomena within real-world contexts. Such is the situation here, with the mobility hub concept still a relatively new development in the transport planning and engineering fields, and a lack of established practices or understandings. A challenge in using case-based approaches, however, is the need to be seeking generalisable findings despite researching at depth within the contexts of the individual case or cases. This depth is possible through including only a small number of cases in the study, typically up to five to ten but sometimes as few as one, yet leads to the suggestion that researchers should explictly respond to this "duality criterion" Benbasat et al. (1987); Eisenhardt (1989); Dyer et al. (1991); Cavaye (1996); Darke et al. (1998); Meredith (1998); Stuart et al. (2002); Voss et al. (2002); Denscombe (2007); Eisenhardt and Graebner (2007); Yin (2009); Ketokivi and Choi (2014); Yin (2014, 2018).

Here the need to be seeking generalisable findings is addressed directly by the form taken by the research outputs: namely, the research produces a software of package tools that might be used for mobility hubs in any location, not just the locations selected as cases. More generally, the case study (i.e. field visits) component of this research is only part of the diverging 'discovery' and 'development' phases, with additional examples of mobility hubs examined through review of websites, online imagery of street locations and other means. As such, while there may be other approaches to information provision, mapping and branding at mobility hubs not examined in this study, it is expected that the study has enough 'breadth' to offset the possible limitations to generalisability that might come from examining only a small number of cases a great 'depth'.

Cases selected for this study was somewhat constrained by circumstances and opportunity, at least for those that were visited in person. The logisitics of and avaiable time for travel, unfortunately, limited case selection to those that were close to the researchers' home base in Ireland or that could otherwise be visited as part of other travel. Hence, most of the visited sites were in the south of England, including in the vicinity of London, Oxford, Birmingham and Bristol. However, and in line with the recommendations of the methodological literature, the cases selected for site visits include those that are leading and particularly novel examples, those that are more representative, and sufficient variation to provide for cross-case comparison. More practically though, the actual conditions at each hub were only briefly examined before each site visit, so as to retain some level of randomness in the selection as far as the inclusion of good, average and not as good examples of mobility hubs in the sampled sites. The location of the mobility hubs themselves was sourced from (INSERT CITATION TO THE MOBILITY HUB WEBSITE), which includes hundreds of sites across the UK and Europe.

In the 'definition' phase the findings from site visits and other exploration of the cases are synthesised through cross-case comparison so as to arrive at a design problem definition. The 'diverging' development phase is then guided by the standardise approaches to package development for the R language, using **?** as a key reference. While there are many other computing languages and software tools that might have been selected instead, R is chosen here because: it is open source and freely available, therefore supporting the goals of this research that relate to developing an accessible suite of tools for practitioners, advocates, tactical urbanists and others who might not have immediate access to commercial software; it has an extensive set of packages for geospatial analysis and mapping, including *sf* (Pebesma, 2026), and the *ggspatial* and *osmdata* (**?**, **?**) packages that facilitate access to Open Street Maps (OSM) datasets and mapping tiles; and there is a large community of users, the availability of beginner learning and reference material[5] and an online culture that appears to support those with less experience with troubleshooting code examples. This community of users is also responsible for the development and maintenance of a wide range of R packages, including: the *tidyverse* (Wickham et al., 2019); *gtfstools* (**?**); *tidytransit* (**?**); and INSERT OTHERS HERE, which were all relied upon to support the code development aspects of this research.

4. Results

4.1. Discovery

4.2. Definition

4.3. Development

4.4. Delivery

5. Discussion

6. Conclusions

[5]See for example INSERT THE MONASH COURSES HERE, AND THE O'REILLY ANIMAL BOOKS

Table 1: Author contribution matrix, as per CRediT (Contribution Roles Taxonomy

Component	NA	BC	BM	SM	JR	AS	AV
Conceptualization					X		
Data Curation					X		
Formal Analysis					X		
Funding Acquisition		X				X	
Investigation					X		
Methodology					X		
Project Administration				X			
Resources							
Software					X		
Supervision			X				X
Validation							
Visualization					X		
Writing – Original Draft Preparation					X		
Writing – Review & Editing							

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Appendicies

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